1. Clinical Question & Core Hypothesis

Clinical Question

mTORC1 inhibition (rapamycin) triggers autophagy, which can either drive tumor cell death or support tumor survival/metastasis; why does this lead to improved overall survival in some tumors but not others? 2) Tier 1 paths used: (a) Rapamycin → mTORC1 (pharmacologic lever); (b) mTORC1 inhibition → autophagy → cell death; (c) cell death → induces TP53 (p53 as a node that interprets death/autophagy signals). These provide a direct mechanistic chain linking drug action to autophagy outcomes that are modulated by TP53. 3) Tier 2 evidence supporting the claim: associations between TP53 status and prognosis, TP53 mutation enrichment in high■risk cohorts, and observed links between metastasis and worse OS motivate testing TP53 as a moderator of whether autophagy results in lethal versus pro-survival programs; reported rapamycin sensitivity in some high■risk cohorts supports a therapeutic angle. 4) Why OS as the endpoint: OS is the most robust, broadly available clinical readout in the provided dataset and directly linked in the supporting paths to metastasis and TP53 prognostic effects. 5) Deliberate exclusions: I excluded Tier 3 exploratory items (hematogenous dissemination, diabetes, KRAS, perinephric abscess, etc.) from the core hypothesis to keep the question mechanistic and testable; additional contextual modifiers (tumor type e.g., bladder cancer) are placed in secondary_modifiers for follow-up rather than the central claim?

Working Hypothesis

In TCGA-style breast and bladder cancer cohorts, high stromal-to-tumor ratio (TSR_high) creates a microenvironment that modulates autophagy in response to mTORC1 inhibition. The effect of rapamycin-induced mTORC1 inhibition on tumor cell fate is TP53-status dependent: TP53 wild-type (TP53 WT) tumors channel autophagy toward tumor cell death and reduced metastatic potential, producing an improvement in overall survival (OS) with mTORC1 inhibition; TP53-mutant (TP53 MUT) tumors rewire autophagy to support survival and metastasis, yielding attenuated or worsened OS under mTORC1 inhibition. Therefore, TSR_high/TP53 WT tumors are predicted to show improved OS when treated with rapalogs compared with TSR_high/TP53 MUT tumors. TSR_low tumors are expected to show a distinct or attenuated response, serving as a relevant comparator. The central testable claim: there exists a significant interaction between TSR, TP53 status, and mTORC1 inhibition such that TSR_high/TP53 WT tumors exhibit a measurable OS benefit from rapamycin exposure relative to TSR_high/TP53 MUT tumors, with the effect on OS not being uniform across TSR strata.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	mTORC1, TP53
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

mTORC1: mTORC1 inhibition (rapamycin) triggers autophagy, which can either drive tumor cell death or support tumor survival/metastasis; why does this lead to improved overall survival in some tumors but not others? 2) Tier 1 paths used: (a) Rapamycin → mTORC1 (pharmacologic lever); (b) mTORC1 inhibition → autophagy → cell death; (c) cell death → induces TP53 (p53 as a node that interprets death/autophagy signals).

TP53: TP53 status and prognosis, TP53 mutation enrichment in high■risk cohorts, and observed links between metastasis and worse OS motivate testing TP53 as a moderator of whether autophagy results in lethal versus pro-survival programs; reported rapamycin sensitivity in some high■risk cohorts supports a therapeutic angle.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	rapamycin	Supporting modifier from knowledge graph
Secondary	TP53_mutation_status	Supporting modifier from knowledge graph
Secondary	metastasis	Supporting modifier from knowledge graph
Secondary	bladder_cancer_context	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Direct evidence in the specific cancer type may be limited; the hypothesis extends findings from related contexts.

Alternative Directions

- rapamycin: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TP53_mutation_status: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.